

(Toward)

Safer Agent Infrastructure

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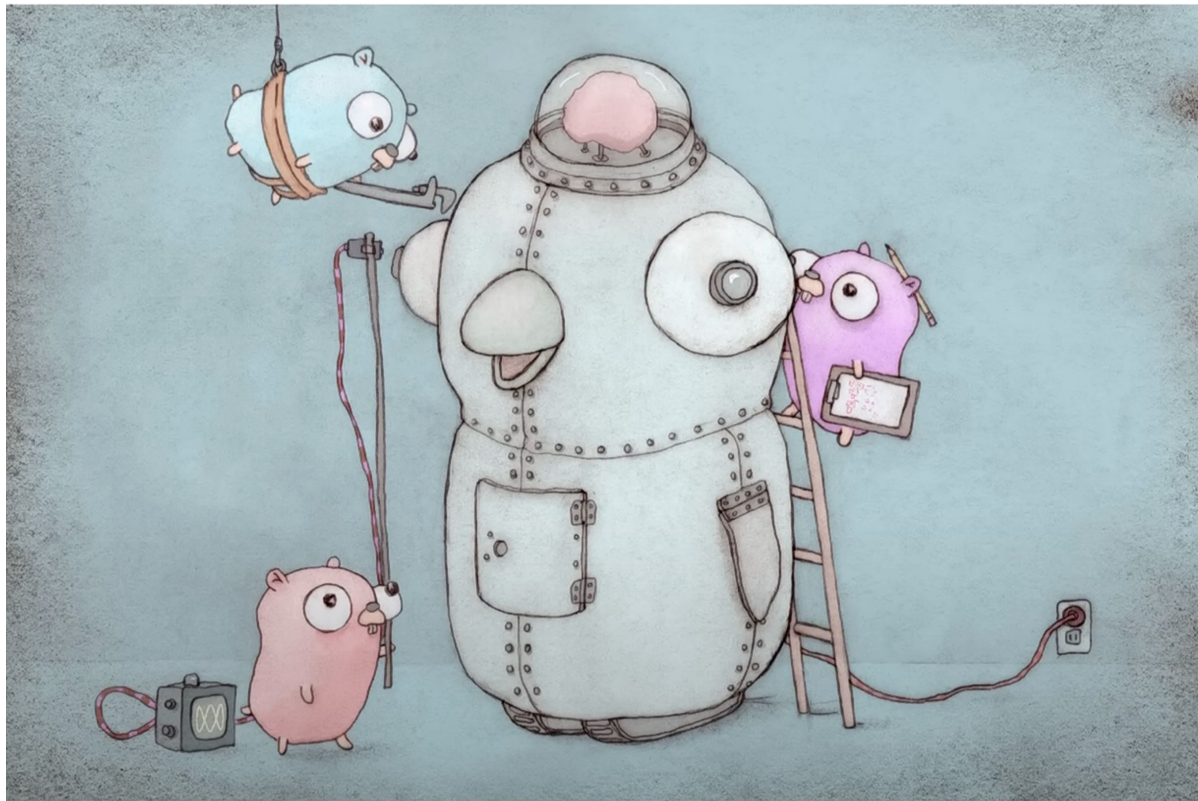
Developer Relations Engineer,
Google Cloud



This is about agents you build



ADK
Genkit
Langchain-go

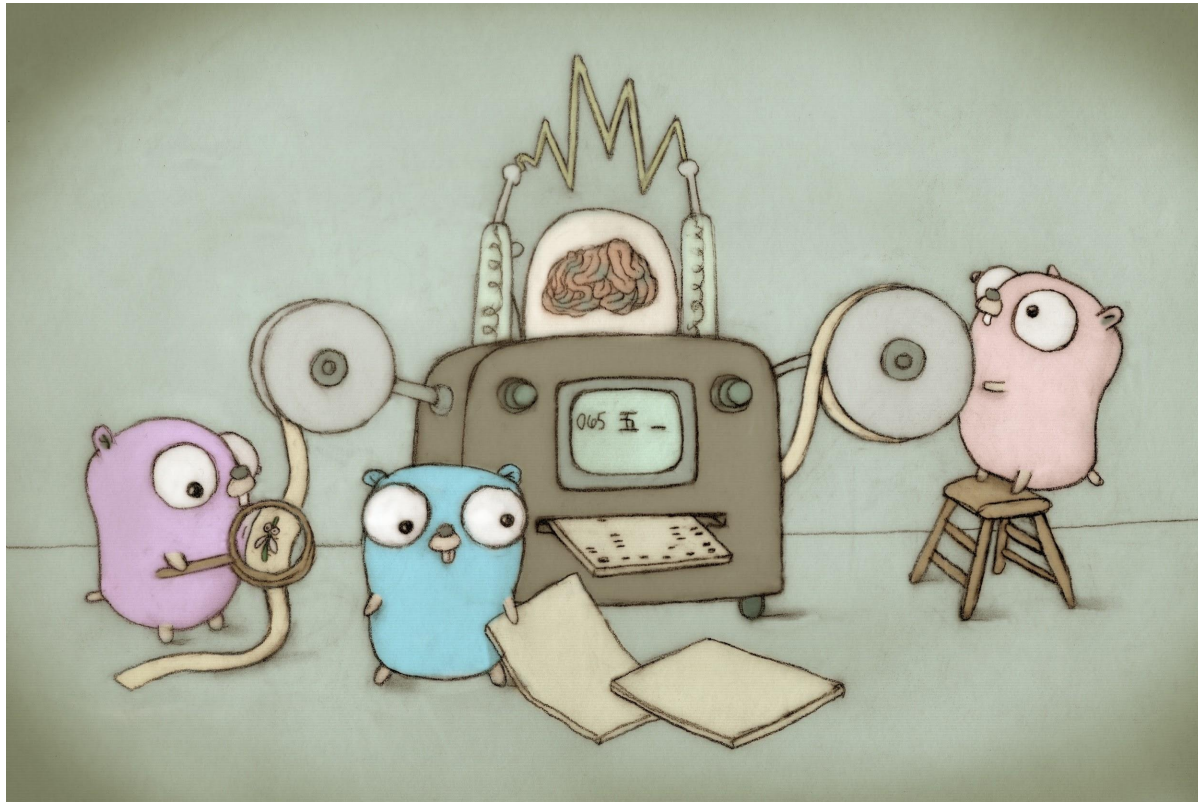


Agents are pretty useful...



But also...

Highly personal
Acting as you
Run as desktop software



Risk: Agents use Ambient Developer Credentials



- No visible difference between agent activity and human activity
- Agents might leak or misuse your access!

- Writing when you wanted a read
- "Summarize this issue for me"
 -  "I've posted a summary to the issue!"
- "Draft an agenda for this meeting"
 -  "I've send the agenda"



Risk: Auth token leaks



- Key exposure has always been a problem
- The consequences are now magnified:
 - Autonomous agents can misuse keys **a lot**
 - Many AI/Cloud services have poor abuse controls



Auth token injection with `http.RoundTripper`



```
// AuthRoundTripper injects an auth header
type AuthRoundTripper struct {
    http.RoundTripper
    HeaderName  string
    HeaderValue string
}
func (rt *AuthRoundTripper) RoundTrip(req *http.Request)
(*http.Response, error) {
    clone := req.Clone(req.Context())
    clone.Header.Set(rt.HeaderName, rt.HeaderValue)
    return rt.RoundTripper.RoundTrip(clone)
}
```

RoundTripper usage



```
// Get Auth token from somewhere
cred := GetAuthCredentials(ctx)

// Use in calling APIs directly or with client library
c := &http.Client{
    Transport: &AuthRoundTripper{
        HeaderName:  "Authorization",
        HeaderValue: "Bearer "+string(cred),
        RoundTripper: http.DefaultTransport,
    }
}
srv, err := calendar.NewService(ctx,
    option.WithHTTPClient(client))
```

“

Time for a thought
experiment!

”

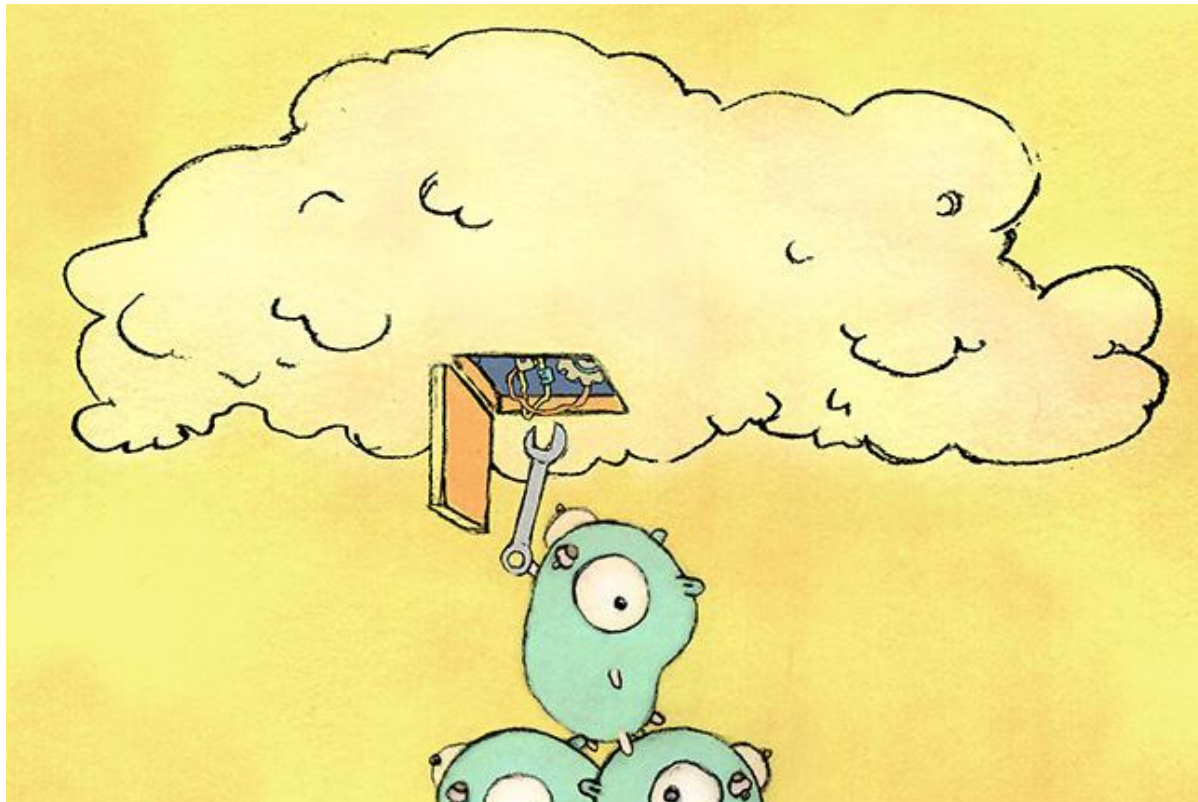


What if we built agents as cloud services?

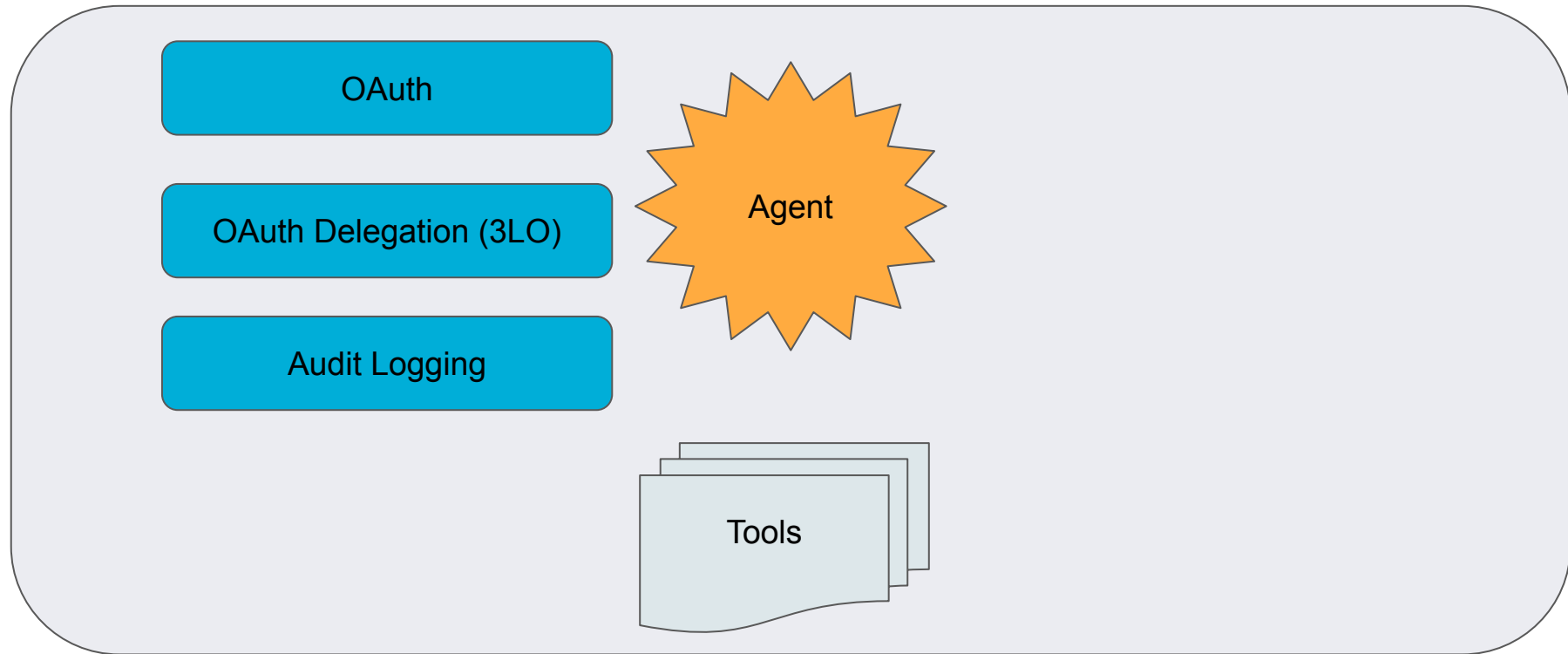


Why?

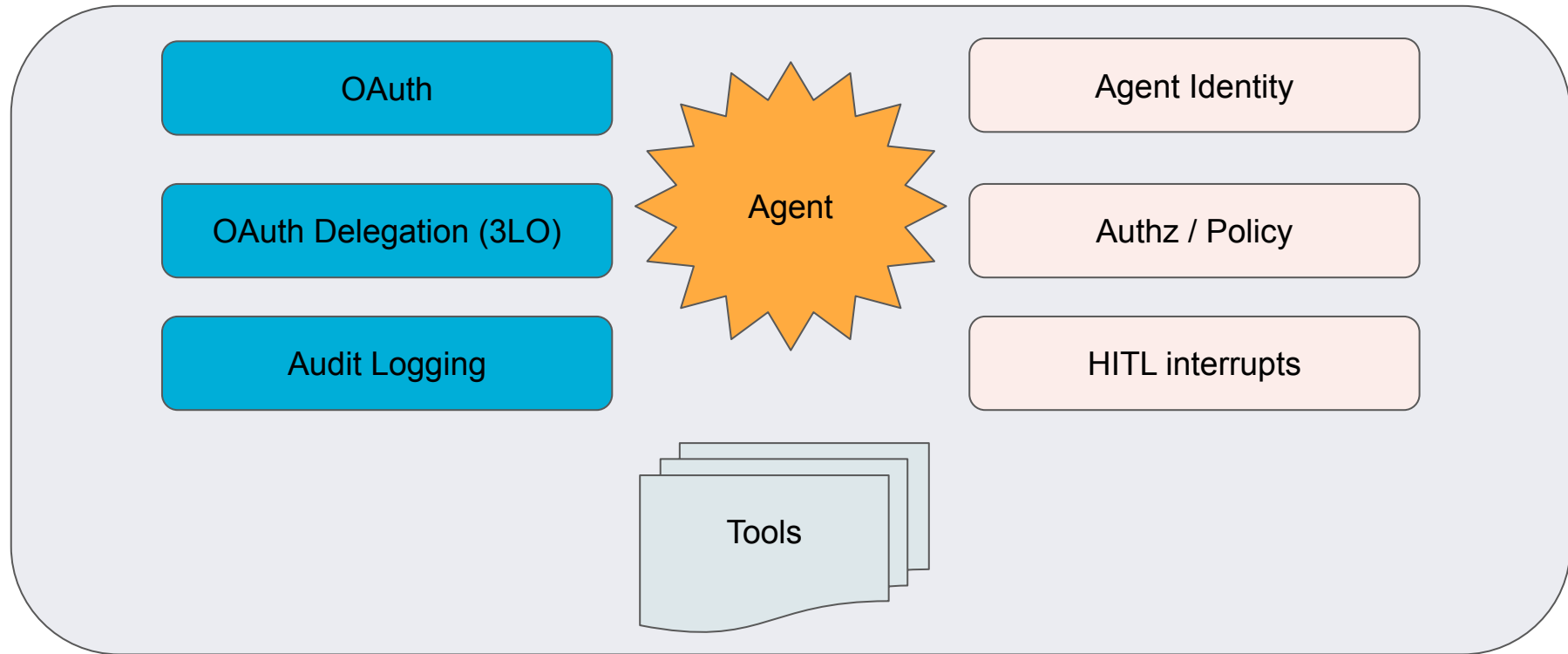
- Persistent
- Multi-user
- More consistent behavior
- Improve access controls
- Audit Logging
- Quota management



Many controls are familiar

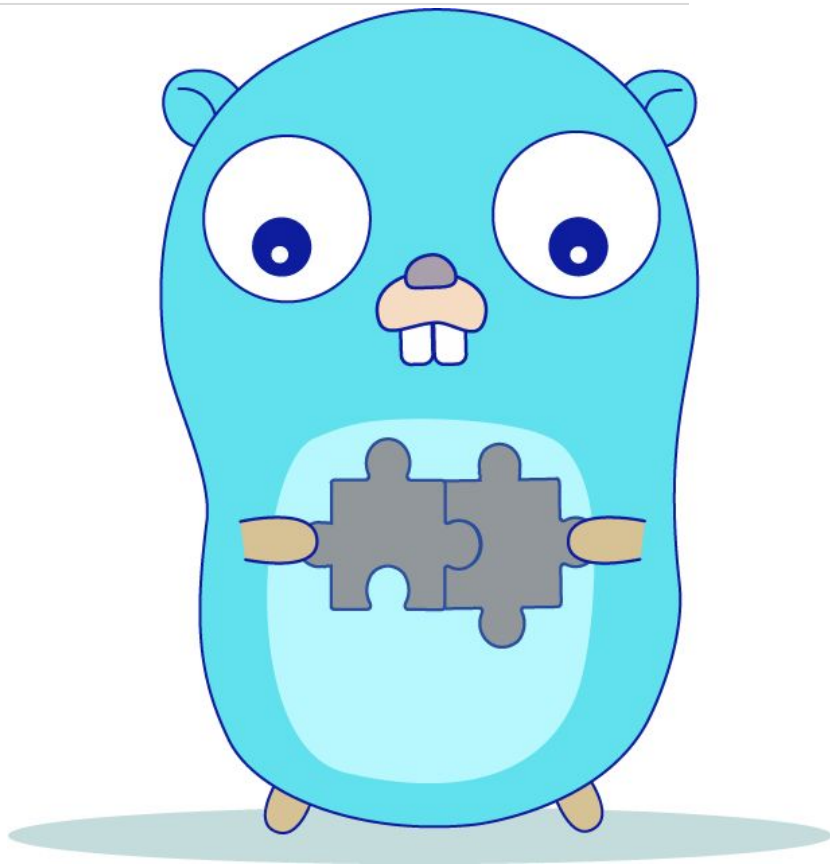


And there are some new pieces



What's missing?

- Authentication / Identity
- Delegation chains
- Quota management
- ???



AgentHost: experiments in agent maturity



<https://github.com/GoogleDevRelExplorations/agenthost>

This is highly experimental!

